4.2. Lipid-Lowering Drugs

Patients with RA have up to twice the risk of atherosclerotic cardiovascular disease seen in the general population, a level similar to that in diabetes [1], which makes lipid-lowering drugs a logical choice to test. The two classes examined here, statins and fenofibrate, also affect immune cells and may benefit both the joints and the heart (Sections 4.2.1 and 4.2.2).

4.2.1. Statins

Statins act mainly as competitive inhibitors of 3-hydroxy-3-methylglutaryl-CoA reductase (HMG-CoA reductase), binding the enzyme at nanomolar concentrations and displacing its natural substrate, which blocks de novo cholesterol synthesis [2]. Statins also have effects beyond lipid lowering, because fewer isoprenoid intermediates leave small guanosine triphosphatases (GTPases) such as Ras, Rho, Rac and Cdc42 less prenylated and therefore less active [2]. They also block interferon-γ (IFN-γ)-induced major histocompatibility complex (MHC) class II expression and thus MHC-dependent T-cell activation [3], and T cells form the largest lymphocyte population in rheumatoid synovium [4].

Animal data on statins are conflicting. In a preclinical study, mice with serum-induced arthritis were given atorvastatin, pravastatin or simvastatin intravenously (0.2 mg/kg on days 3, 5 and 7), and atorvastatin and pravastatin lessened clinical arthritis and preserved joint structure on paw histology, but simvastatin had no significant effect [5]. This regimen corresponds to approximately 1 mg per dose in a 60-kg adult after body-surface-area conversion [6], far below the 40 mg/day used in RA trials [7, 8], although intravenous and oral doses are not directly comparable.

An earlier CIA study with simvastatin was favorable. In mice given daily i.p. simvastatin at 10, 20 or 40 mg/kg between days 21 and 40, arthritis was suppressed in a dose-dependent manner [9], and a 14-day course of 40 mg/kg, started one day after clinical onset, also inhibited established disease [9]. Because these doses did not significantly lower plasma cholesterol, the effect appeared largely independent of lipid lowering [9]. Findings with other statins were less favorable. In a later CIA study, atorvastatin and rosuvastatin had no effect on clinical, histologic, immunologic or biochemical markers of disease

[10]. The benefit of i.p. simvastatin at 40 mg/kg/day, the dose used by Leung et al., was reproduced, but the authors attributed it to severe toxicity with raised glucocorticoid levels [10]. Deaths reduced this group from 18 to 7 mice over the study, which suggests that the earlier benefit may reflect toxicity rather than a true antiarthritic effect [10]. When Vandebriel et al. gave CIA mice oral atorvastatin at 0.2 mg/day or oral pravastatin at 0.8 mg/day, arthritis started earlier, and every treated mouse became arthritic, against 7 of 12 controls [11]. After body-surface-area conversion for a 60-kg adult [6], the i.p. simvastatin doses of 10–40 mg/kg [9] correspond to approximately 49–195 mg/day, with the highest dose about five times the 40 mg/day used in RA trials [7, 8]. The oral atorvastatin dose used by Vandebriel et al., about 10 mg/kg, corresponds to approximately 49 mg/day and is close to the trial dose, whereas pravastatin at four times that dose corresponds to approximately 195 mg/day [11]. Earlier arthritis onset after statin treatment has, to our knowledge, been reported only by this group [11].

Results in patients are mixed. Two meta-analyses showed that statins reduced disease activity measured by DAS28, with standardized mean

differences (SMDs) of −0.55 and −0.79 (95% CIs −0.83 to −0.26 and −1.08 to −0.49, respectively) [12, 13]. However, the included trials were small and mostly conducted at single centers [12], and both meta-analyses reported high heterogeneity [12, 13]. The larger one also found possible publication bias for DAS28 (Egger test P = 0.012) [13]. In an unblinded randomized study conducted in Egypt, 30 patients with early RA were divided into two groups of 15 and treated for 6 months with MTX (0.2 mg/kg/week) and prednisone (10 mg/day), either alone (group 1) or with atorvastatin 40 mg/day (group 2). DAS28 was significantly reduced from 6.09 to 3.9 in group 2 compared with 6.19 to 5.27 in group 1 [14]. The double-blind Trial of Atorvastatin in Rheumatoid Arthritis (TARA) randomized 116 patients to atorvastatin 40 mg/day or placebo added to their existing DMARDs [7], with co-primary outcomes of change in DAS28 and European League Against Rheumatism (EULAR) response [7]. After 6 months, DAS28 improved more with atorvastatin than with placebo (between-group difference −0.52, P = 0.004; 95% CI from −0.87 to −0.17) [7]. This difference was slightly below the 0.6-unit change the trial was powered to detect, but

EULAR response was more frequent with atorvastatin (31% vs 10%) [7], whereas the small unblinded Egyptian study reported a between-group difference of about 1.3 units [14]. In a 12-week randomized study without placebo (blinding not reported), 55 patients with RA received atorvastatin 20 mg/day or no statin [15]. DAS28 fell by 0.41 with atorvastatin and by 0.03 without it ($P < 0.001$), and CRP and ESR also decreased [15]. The DAS28 change, however, was below the 0.6-unit threshold used to power TARA [7]. In contrast, the largest trial we found, the double-blind Trial of Atorvastatin for primary prevention of Cardiovascular Events in RA (TRACE RA), enrolled 3,002 UK patients for cardiovascular end points [8]. It found no difference in DAS28 between the atorvastatin and placebo groups [8]. Atorvastatin did lower CRP, but the authors judged the fall too small to matter clinically [8]. TRACE RA targeted cardiovascular end points, reported compliance in its atorvastatin arm fell from 89% at 3 months to 39% by 60 months, and the trial was stopped early because cardiovascular events were fewer than expected [8]. These design features, rather than a lack of efficacy alone, may partly explain why TRACE

RA did not confirm the smaller trials.

Observational data on statin use and RA onset are inconsistent: in an Israeli cohort of 211,627 new statin users, persistence was measured using the proportion of days covered (PDC) [16]. Compared with a PDC of <20%, a PDC of ≥80% was associated with lower RA risk (adjusted HR 0.58, 95% CI from 0.52 to 0.65), and to test for healthy-user bias, the same study also examined osteoarthritis [16]. For osteoarthritis, high persistence was associated with only a modest risk reduction (HR 0.85, 95% CI from 0.81 to 0.88) [16]. A US case–control study that matched 16,363 RA cases with the same number of controls initially found a slightly higher risk with statin use [17], but after adjustment for hyperlipidemia the association disappeared, with OR 0.95 (95% CI from 0.90 to 1.01) [17]. However, hyperlipidemia is often diagnosed when statin treatment begins, which may limit the accuracy of this adjustment [17]. Genetic data, in contrast, suggest a protective role. In a drug-target Mendelian randomization study, genetically proxied HMG-CoA reductase inhibition was linked to lower RA risk in both the discovery (OR 0.45, 95% CI from 0.32 to 0.63) and replication (OR 0.37, 95% CI from 0.23 to

0.61) datasets [18]. Although consistent, these data come from European populations only and address RA onset rather than disease activity in established RA [18]. A small prevention trial was also inconclusive: among 62 anti-citrullinated protein antibody (ACPA)-positive patients with arthralgia, atorvastatin 40 mg/day was compared with placebo for a planned 3 years, but the trial was stopped early because of low inclusion [19]. Arthritis developed in 9/31 (29%) atorvastatin recipients and 6/31 (19%) placebo recipients (HR 1.40, 95% CI from 0.50 to 3.95), a non-significant difference [19]. Only 62 of the planned 220 participants were enrolled, which left the trial too small to confirm or exclude a protective effect [19]. For now, statins in RA are prescribed to lower lipids and cardiovascular risk, as in other patients [8], and larger, longer placebo-controlled trials with disease activity as the primary end point are needed before statins can be used to treat RA itself.

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